Conveyor + Camera Control Panel

Rebuild guide, Revision 5 (13 October 2026): panel, PLC, HMI, VFD, camera and PC program.
Based on the old TIA Portal project "latest plc", your hand drawings, your BOM, your explanation of how the machine works, and the PC and camera files in the folder conveyor_automation-main.

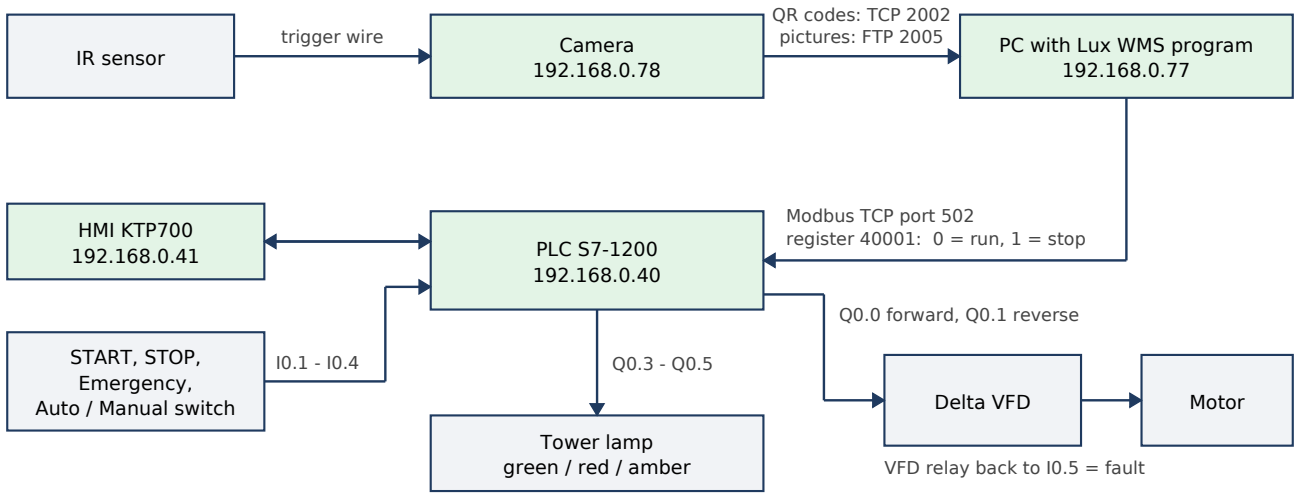
What changed in Revision 5. The VFD part is rewritten with exact facts in place of "copy from the old drive" and "check": section 7 now says what every important setting of the old drive means (from the Delta VFD-EL manual), which wire goes to which terminal and why, the input mode switch (PNP), where the speed comes from (the knob on the drive), and the list of parameters to set. New **Appendix C** lists every non-zero setting of the old drive with its name, value and factory value. Sections 2, 3, 4 and 6 are updated to match.

What changed in Revision 4. On 12 October 2026 the old project (a copy) was opened in TIA Portal and the whole PLC program and HMI were exported by TIA itself. The program description in this guide is now checked against that export (Appendix A). Also:
1. The red-button reverse in MANUAL is **not** in the old program; you confirmed it was wanted but never built. It is now a planned new feature with one extra wire to spare input I0.7 (sections 2, 3, 4.1, 6.2, 13; how-to guide part 12).
2. HMI notes in section 11 (the BACK button on OUTPUT STATUS does nothing in the old project).

What changed in Revision 3. Section 7 now shows the real values read from the old drive on 7 October 2026 (motor current 07.00 corrected: the old drive holds 2.5 A, not 1.82 A), and the VFD tool notes say it has been used on the real drive. Files are now in numbered folders (see README.md in the PLC folder).

What changed from Revision 1.
1. The IR sensor is wired to the **camera**, not to the PLC. PLC input I0.0 stays empty. Revision 1 told you to wire the sensor to I0.0. That was wrong.
2. The conveyor is stopped and started by the **PC program** through Modbus register 40001. This is the real design. Revision 1 called it a test left in the program. That was wrong too.
3. The PC program and two camera setting files were found in your folder. New sections 8 and 12 describe them.
4. The VFD relay wiring (I0.5, purple wire, health lamp) is now read from your sketch (section 4.2).
5. New: three ways to copy the VFD parameters, with a ready tool (section 7).

1. How the machine works



AUTO (selector at AUTO)

1. Emergency released, no fault. Press the green **START** button. The PLC remembers the start ("ACTIVE"), the green lamp is ON and the conveyor runs forward. It runs as long as register 40001 is 0.
2. The operator puts a row of boxes on the belt. The IR sensor sees them and triggers the camera **directly** (the PLC is not involved).
3. The camera waits for its trigger delay (3 seconds in the saved settings), so the boxes are in the middle. It takes one picture, reads up to 6 QR codes, and sends the codes and the picture to the PC.
4. The PC program counts the codes. Count equal to the **Stack Size** chosen on the PC screen (6 when the program starts; you can pick 2, 4, 6, 8, 10, 20 or type a number): nothing happens, the conveyor keeps running. Count different: the PC writes **1** to register 40001 and the PLC stops the conveyor.
5. The operator corrects the boxes and clicks **Rescan Stack** (or **Start**) on the PC screen. The PC writes **0** and the conveyor runs again.
6. The red **STOP** button or the Emergency button stops the cycle at any time.

MANUAL (selector at MANUAL, amber lamp ON)

- Open the **MANUAL** screen on the HMI. Hold **CONVEYOR FWD JOG**, or hold the green START button: the conveyor runs forward while you hold.
- Hold **CONVEYOR REV JOG** on the HMI: reverse while you hold.
- The PC program cannot run the conveyor in manual mode.

After an Emergency stop or a VFD fault

The alarm stays stored. Release the Emergency button (or clear the VFD fault), then press the yellow **FAULT RESET** button on the HMI (bottom right on every screen). Only then will the machine start again.

2. Result of my verification

Item	Result	What it means for you
Your explanation of how the machine works	OK	It matches the PLC program (exported by TIA itself on 12 Oct), the PC program and the camera settings. The red button reversing in manual mode is not in the program: you confirmed it was wanted but never built. It is now a new feature (section 6.2, how-to guide part 12).
PLC and HMI part numbers in the BOM	OK	Exactly the same as in the old program (CPU 1214C DC/DC/DC 6ES7214-1AG40-0XB0, HMI KTP700 Basic 6AV2123-2GB03-0AX0).
Power drawing (page dated 18 Aug)	OK	Matches the BOM. One wording fix: the "DC-DC converter" is really the 230 VAC to 24 VDC power supply.
PLC I/O drawing (page dated 19 Aug)	OK	Every signal matches the program. I0.0 with no wire is correct, because the sensor goes to the camera.
VFD relay, purple wire, health lamp	OK	Your sketch: RC = +24 V, RA to I0.5 and the purple lamp wire, RB to the health lamp. The old drive has the relay set to "fault indication" (03.00 = 8), which is exactly what this wiring needs (section 7).
Q0.2 (fault reset wire to the VFD)	CHECK	The program switches Q0.2, but your sketch shows no wire. The VFD-EL-W has only inputs MI1 - MI3, and the old drive's free input MI3 has no function (04.05 = 0), so a Q0.2 wire would do nothing: leave Q0.2 unwired, as on the old panel. A drive fault is reset with the drive's STOP/RESET key.
PC program (Lux WMS)	OK - found	It is in conveyor_automation-main / Frontend. Three Python packages are used but not listed in its install file (section 12).
Camera settings	CHECK	Two setting files are in your folder (saved 19 and 23 Feb 2026). The picture-upload port inside them (21 / 8080) is not the port the PC program uses (2005), so they are close but not the final settings. Best: export the settings again from the old camera (section 8).
VFD parameter settings	OK - read	All 306 parameters were read from the old drive on 7 Oct 2026. Their meaning and the list to set are in section 7 and Appendix C.
Power for camera and light	CHECK	Not on your drawing and "not confirmed" in the BOM. Find how the old camera and light get 24 V.

Safety. The panel contains 230 VAC. Mains wiring, the VFD and the motor must be wired and checked by a qualified electrician, with the main isolator OFF and locked. In this design the Emergency Stop only goes to a PLC input (software stop). It does not cut the power to the VFD by itself.

3. Do this on the OLD machine before you buy or dismantle anything

#	What	How	Why
1	Reverse test in manual mode: **settled**	You confirmed on 12 Oct 2026: the red button does **not** reverse on the old machine; it was wanted but never built. So the file "latest plc" matches the machine. The reverse jog is added as a new feature (section 6.2).	Nothing left to test here.
2	VFD: two things to look at	The parameters are read already (7 Oct). Look only at: (a) the small NPN / PNP switch on the old drive (it should be at PNP), (b) whether any wire goes from PLC Q0.2 to the drive (expected: none).	Section 7.2 explains both.
3	Camera	Connect the HIKROBOT IDMVS software to the old camera and export its settings to a file. Photograph the camera cable: which wire carries the sensor signal, and where the camera and the light get their power.	The saved setting files are a few days older than the final machine.
4	Sensor, tower lamp, buttons	Photograph the labels and the back of each button. Count the NO and NC contact blocks.	The BOM lists these as "type only". STOP and Emergency must use NC contacts.
5	Wire-by-wire photos	Every PLC terminal, the VFD control terminals (RA, RB, RC, MI1, MI2, DCM, 24V) and the terminal strips.	To settle the CHECK points of section 2.

Best extra protection: do **Online > Upload device as new station** once on the old PLC and keep the result. It is the only copy that is certain to be what the old machine runs.

4. Check of your hand drawings against the program

4.1 PLC inputs and outputs

PLC terminal	Your drawing	Program	Result
I0.0	Nothing	Tag "DI_PRODUCT SENSOR" exists, but its logic is switched off	OK - leave empty. The sensor goes to the camera.
I0.1	Selector SW (AT)	Auto / Manual selector. ON = AUTO, OFF = MANUAL	OK
I0.2	PB Green	Cycle START push button (NO contact)	OK
I0.3	PB Red	Cycle STOP push button (NC contact)	OK - confirm the contact is NC
I0.4	Em switch	Emergency stop (NC contact)	OK - confirm the contact is NC
I0.5	To VFD relay RA	VFD fault. Input ON = fault	OK
I0.6	Joined to the I0.5 / RA wire, with the purple lamp wire	Not used	OK - the screw is only used as a joint. Copy it.
I0.7	Nothing	Spare (tag DI_SPARE I0.7)	OK. Gets a new wire only if you add the red-button reverse (section 6.2).
Q0.0	To VFD	Conveyor FORWARD	OK
Q0.1	To VFD	Conveyor REVERSE	OK
Q0.2	Nothing	Fault RESET signal to the VFD	OK - leave unwired: the VFD-EL-W has only MI1 - MI3 and MI3 has no function (04.05 = 0). So the HMI reset clears the PLC alarm only, and the VFD must be reset at the drive.
Q0.3	Lamp green	Auto cycle started	OK
Q0.4	Lamp red	Fault	OK
Q0.5	Lamp yellow	Manual mode	OK
Q0.6	Crossed out	Not used	OK
Input common (top "0V")	0 V	1M = 0 V, because the switches give +24 V	OK
Output supply ("+ -")	+24 V and 0 V	3L+ = +24 V, 3M = 0 V	OK

4.2 VFD relay, as I read your sketch

VFD terminal	Goes to	What it does
RC (common)	+24 V	Feeds the relay contact
RA (closes when the relay pulls in)	PLC I0.5, with a link to the I0.6 screw, where the purple wire of the tower lamp is also connected	On a VFD fault: the PLC gets the fault signal and the purple wire (buzzer or fourth light) gets 24 V
RB (closed while the relay is at rest)	The "VFD healthy" indicator lamp on the panel door (other lamp wire to 0 V)	Lamp ON while the drive has no fault

The old drive has parameter **03.00 = 8 = fault indication** (read on 7 Oct 2026; meaning from the Delta manual). With this setting the relay behaves exactly as the table says, so the sketch and the drive agree. Set 03.00 = 8 on the new drive too (it is also the factory value).

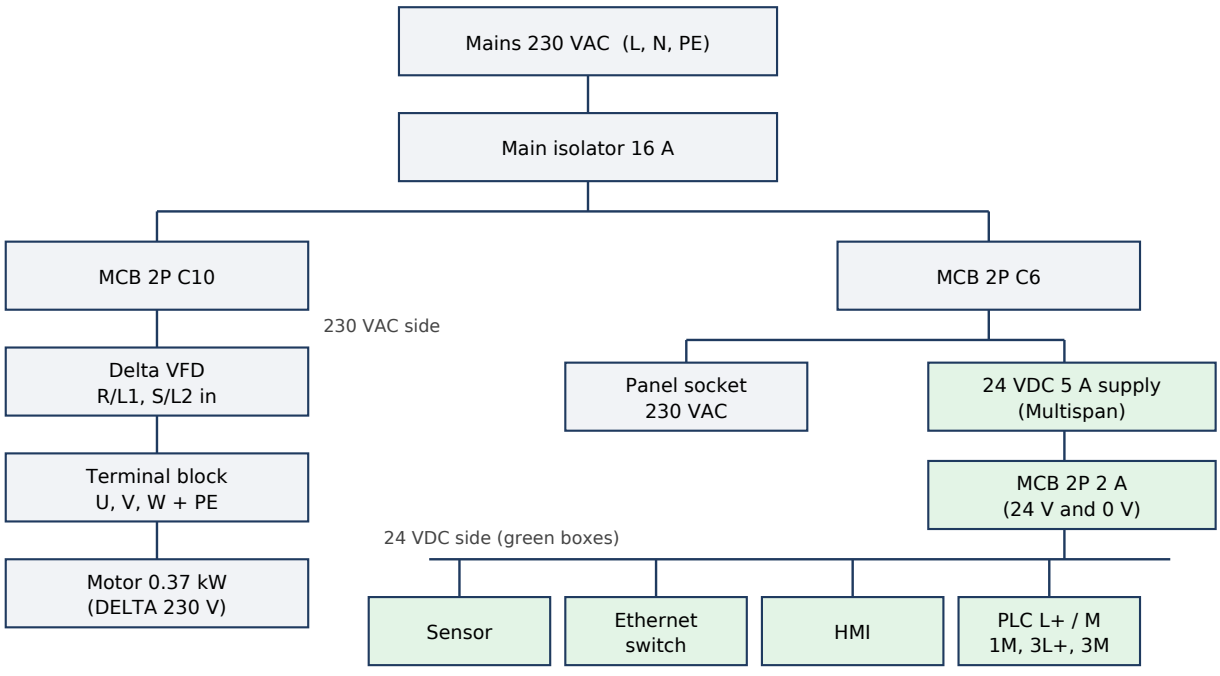
4.3 Power drawing

Your drawing	BOM part	Result
Main power L, N to main switch	Salzer 16 A rotary isolator	OK
MCB C10 to VFD to motor (U, V, W)	Schneider A9N2P10C, Delta VFD004EL21W-1, Bonfiglioli BE 71B 4	OK - the motor terminal box must be linked in DELTA (230 V). See section 7.
MCB C6 to socket and "DC-DC converter"	Schneider A9N2P06CGN, Multispan PS24-5A-P	OK - the converter is the 230 VAC / 24 VDC 5 A power supply
24 V, 0 V to MCB 2A	Schneider A9KF71202	OK
2A MCB to sensor, Ethernet switch, HMI, PLC	Teltonika TSW010, KTP700, CPU 1214C	OK
Camera and light supply	BOM: "not confirmed"	CHECK - not on your drawing

4.4 BOM - items to add or settle

- **Ethernet patch cables**: PLC, HMI and PC to the switch, plus one for the programming laptop. The switch has 5 ports: PLC, HMI, camera, PC and one free.
- **Camera cables**: the camera uses its own round-connector cables for network and for power / signals. Order the same ones as on the old machine.
- **PC for the Lux WMS program**, if the old PC stays with the old machine.
- **TIA Portal V21** (STEP 7 Basic + WinCC Basic). The project cannot be opened with an older version.
- **USB-to-RS485 adapter** (optional, low cost) to read the VFD parameters with the laptop. See section 7.
- **Contact blocks**: START 1 NO, STOP 1 NC, Emergency 1 NC, selector 1 NO.
- **Signal tower**: 24 VDC, red / amber / green plus the fourth (purple) wire, common 0 V.
- **Photoelectric sensor**: same model as the old one if readable. Its output must suit the camera trigger input, so copy the old type.

5. Power wiring



From	To	Note
Mains L, N, PE	Incoming terminals, then isolator	PE (earth) goes to the earth bar and to the mounting plate, VFD, motor, power supply and PLC earth terminals.
Isolator out L, N	MCB C10 in and MCB C6 in	
MCB C10 out	VFD R/L1 and S/L2	Single-phase 230 V input drive.
VFD U/T1, V/T2, W/T3	Terminal block, then motor U, V, W	Use a screened motor cable if the old one is screened.
MCB C6 out	Panel socket, and 24 V supply L, N	
24 V supply +V, -V	MCB 2 A, then the +24 V and 0 V terminal rows	Set the supply output to 24.0 V before connecting loads.
+24 V / 0 V terminals	PLC, HMI, Ethernet switch, sensor, button common, VFD relay RC, health lamp	See the next sections for the exact terminals.
Camera and light	Copy from the old machine	The camera needs 12 - 24 VDC (up to 12 W). The light has its own rating: read its label.

6. PLC wiring (CPU 1214C DC/DC/DC)

6.1 Supply and commons

PLC terminal	Connect to	Purpose
L+	+24 V	CPU supply
M	0 V	CPU supply
Earth terminal	PE bar	Functional earth
1M (input common)	0 V	Inputs turn ON when +24 V arrives on them
3L+	+24 V	Supply for the outputs
3M	0 V	Output reference

6.2 Inputs (top terminal row)

Input	Device	Contact	Wiring	Input is ON when
I0.0	Not connected	-	Leave empty	-
I0.1	Auto / Manual selector	NO	+24 V - contact - I0.1	Selector is at AUTO
I0.2	Cycle START (green)	NO	+24 V - contact - I0.2	Button pressed
I0.3	Cycle STOP (red)	NC	+24 V - contact - I0.3	Button NOT pressed (normal)
I0.4	Emergency stop	NC	+24 V - contact - I0.4	Emergency NOT pressed (normal)
I0.5	VFD fault relay	RA - RC	+24 V - RC, RA - I0.5	VFD has a fault
I0.6	Joint only	-	Wire link from I0.5, plus the purple lamp wire	(not used by the program)
I0.7	**New, optional:** red button, one extra NO contact block	NO	+24 V - new contact - I0.7	Red button held (reverse jog in MANUAL; how-to guide part 12)

With normal healthy conditions the PLC input LEDs must show: I0.3 ON, I0.4 ON, I0.5 OFF. If I0.3 or I0.4 is OFF the machine will never start. If I0.5 is ON the PLC shows VFD FAULT.

6.3 Outputs (bottom terminal row). Each output gives +24 V when ON

Output	Goes to	Function
Q0.0	VFD forward run input (MI1 on Delta EL)	Conveyor forward (auto and manual)
Q0.1	VFD reverse run input (MI2 on Delta EL)	Conveyor reverse (manual only)
Q0.2	**Not connected** (the VFD-EL-W has no input set to reset)	ON while the HMI FAULT RESET button is pressed; resets only the PLC alarm
Q0.3	Signal tower GREEN	Auto cycle started
Q0.4	Signal tower RED	Fault present
Q0.5	Signal tower AMBER / yellow	Manual mode selected
Lamp common	0 V	

7. VFD and motor

This section is now exact. The values were read from the old drive on 7 October 2026 (all 306 parameters), and their meanings are taken from the **Delta VFD-EL user manual** (parameter summary, chapter 4.1). Your drive is the VFD-EL-W, the same family with the same parameter numbers; it has a few extra parameters that this manual does not list (marked in Appendix C).

7.1 What the old drive is set to do

Question	Answer on the old drive	Parameter
Where does the speed come from?	From the **round knob (potentiometer) on the drive's keypad**. The PLC sends no speed. Range 0 to 70 Hz.	02.00 = 4, 01.00 = 70.00 Hz
Where does run / stop come from?	From the **control terminals** (the PLC wires). The STOP key on the keypad also works.	02.01 = 1
Which terminal does what?	**MI1** = forward while ON, **MI2** = reverse while ON (2-wire mode).	04.04 = 0
Which terminals are used? (VFD-EL-W)	The EL-W has only **MI1, MI2, MI3**. **04.19 = 1** switches on 2-wire control so that MI2 means reverse (with 04.19 = 0, the factory value, MI1 is run/stop only and **reverse does not work**). MI3: no function. 04.06 - 04.08 belong to inputs the EL-W does not have; copy them only to be identical.	04.19 = 1, 04.05 = 0, 04.06 = 0, 04.07 = 5, 04.08 = 14
What does the relay RA / RB / RC do?	**Fault indication**: RA - RC closes on a fault, RB - RC is closed while healthy.	03.00 = 8
How fast does it start and stop?	Start ramp 5.0 s, stop ramp 0.5 s, ramp to stop.	01.09 = 5.0, 01.10 = 0.5, 02.02 = 0
Motor data	50 Hz, 220 V. Motor current setting 2.5 A (the drive's own rating; the motor plate says 1.82 A).	01.01 = 50.00, 01.02 = 220.0, 07.00 = 2.5
Motor overload protection by the drive	**Switched off** (this is also the factory setting).	06.06 = 2
Does it start by itself at power-on or after a fault?	No.	02.05 = 1, 08.15 = 0
Fault history	Empty: no fault was stored. The motor has run 633 minutes in total.	06.08 - 06.12 = 0, 07.10 = 633

7.2 Wiring that follows from these settings

Wire	From	To	Why it is certain
Forward	PLC Q0.0	VFD **MI1**	04.04 = 0: MI1 is forward / stop
Reverse	PLC Q0.1	VFD **MI2**	04.04 = 0: MI2 is reverse / stop
Fault reset	PLC Q0.2	**Nothing** (leave unwired)	No EL-W input is set to reset (MI3: 04.05 = 0). Reset a drive fault with the drive's STOP/RESET key
Input common	Panel 0 V (PLC 3M)	VFD **DCM**	The PLC outputs give +24 V; the current must return to 0 V through DCM
Input mode switch	-	Small switch on the drive at **PNP**	With +24 V signals the inputs only work in PNP mode. The factory position is NPN: move it on the new drive, power OFF
Fault to PLC	+24 V to VFD **RC**	VFD **RA** to PLC I0.5 (and the link to I0.6 with the purple lamp wire)	03.00 = 8: RA - RC closes on a fault
Healthy lamp	VFD **RB**	"VFD healthy" lamp, other lamp wire to 0 V	RB - RC is closed while there is no fault
Power	MCB C10: L, N	VFD R/L1, S/L2	Single-phase 230 V drive
Motor	VFD U, V, W	Motor, connected in **DELTA** (230 V)	Motor 230 / 400 V; the drive gives 3 x 230 V. Rated current in delta 1.82 A

Checked on control box 2 (15 Oct 2026):
1. Confirmed on 15 Oct 2026 (control box 2): with MI1 = Q0.0, MI2 = Q0.1, DCM = 0 V, PNP and **04.19 = 1**, forward and reverse both work. Q0.2 stays unwired.
2. The position of the NPN / PNP switch: PNP (the PLC outputs give +24 V).

7.3 Set up the new drive

1. Power OFF: wire as in 7.2 and move the input mode switch to **PNP**. Leave the motor cable off for the first power-up.
2. Power ON, only the drive (MCB C10). The display lights up and shows no fault code.
3. Read the new drive with the laptop tool and compare it with the old drive (7.4). The comparison lists exactly the parameters that are different.
4. Enter each different value with the keypad: press **ENTER** (the parameter group shows), choose the group with the UP / DOWN keys, **ENTER**, choose the parameter number, **ENTER**, set the value, **ENTER**. The display shows **End** when the value is saved, or **Err** when it is not accepted. **MODE** goes back.
5. Read the new drive once more and compare again. Nothing should be left, except read-only and counter values.
6. Speed: on the **old** drive, with the conveyor stopped, press MODE until the display shows **F** and a number: that is the speed setting of the knob. Turn the knob of the new drive until it shows the same number.
7. Power OFF, connect the motor, power ON, and test the direction with FWD JOG (section 13).

What the comparison will most likely list. These are the old values that differ from the factory values printed in the VFD-EL manual. The EL-W may have slightly different factory values, so the comparison of step 3 decides; this table tells you what to expect and what each one is for.

Parameter	Set to	Factory (manual)	What it is
00.12	1	0	50 Hz base voltage: 1 = 220 V / 380 V
01.00	70.00	60.00	Maximum output frequency (top of the knob range)

Parameter	Set to	Factory (manual)	What it is
01.01	50.00	60.00	Motor rated frequency
01.09	5.0	10.0	Acceleration time, seconds
01.10	0.5	10.0	Deceleration time, seconds
02.00	4	1	Speed from the keypad knob
03.08	3	0	Cooling fan: on when the heat sink is warm
04.05	0	1	MI3: no function
04.06	0	2	MI4: no function
04.07	5	3	MI5 function (no MI5 terminal on the EL-W): only to be identical
04.19	**1**	0	**Start / stop mode: 2-wire, MI2 = reverse. Without it reverse does not work**
04.08	14	4	MI6: external fault input
01.12, 01.13	0.1, 0.1	10.0, 1.0	Second deceleration time and jog acceleration time: not used by this machine, set them only to be identical
02.11, 02.12	50.00, 50.00	60.00, 60.00	Keypad / communication frequency: not used (speed comes from the knob), set them only to be identical
07.08	0.10	0.30	Torque compensation time constant: set it if the comparison lists it

Do not copy: 00.02 (parameter reset: 9 or 10 erases all settings), the read-only values (00.00, 00.01, 00.06, 02.16, 02.17, 03.13, 04.26), the fault records (06.08 - 06.12) and the run-time counter (07.10).

Motor protection, for your electrician to decide: the old drive has the motor overload protection switched off (06.06 = 2) and the motor current at 2.5 A, while the motor plate says 1.82 A. An exact copy keeps this. Better protection would be 06.06 = 0 and 07.00 = 1.8; that is a deliberate change from the old machine.

7.4 Read and compare with the laptop tool

Do not connect the drive to the laptop with an Ethernet cable. The RJ45 socket on the drive looks like a network socket, but it is an **RS-485 serial** port. A laptop network port cannot talk to it, and one pin of that socket carries a supply voltage.

1. Power OFF. Connect the USB-to-RS485 adapter (the Sharvi ISO-B-RS485-A used on 7 October): adapter **A+** to drive terminal **SG+**, adapter **B-** to **SG-**. GND is not connected.
2. Power ON, only the drive. Plug the adapter into the laptop (the Lenovo; the adapter appears as /dev/ttyUSB0).
3. In the PLC folder run:
   `/usr/bin/python3 04_VFD/read_vfd_parameters.py --port /dev/ttyUSB0 --out 04_VFD/new_vfd_parameters.csv`
4. Then:
   `/usr/bin/python3 04_VFD/read_vfd_parameters.py --compare 04_VFD/VFD_OLD_MACHINE/old_vfd_parameters.csv 04_VFD/new_vfd_parameters.csv`
   It prints only the parameters that differ, as raw numbers without the decimal point (5000 = 50.00). Or tell me, and I run both steps and give you the list.

- The tool only reads. It cannot write, start, stop or reset the drive (it refuses to send anything but the Modbus read command).

- It was used on the old drive on 7 October 2026: Modbus ASCII, 9600 baud, 7N2, address 1 (the factory communication settings, 09.00 - 09.04).
- Without the laptop: read every value of the new drive with the keypad and compare it by hand with Appendix C.

8. Sensor and camera

The camera has no program to write. It only has **settings**, and the new camera needs the same settings as the old one.

8.1 Wiring

Wire	Connect to	Note
Sensor supply (+ and -)	Panel +24 V and 0 V (after the 2 A MCB)	As on your power drawing
Sensor output	The trigger input wire of the camera cable	Copy the exact wire from the old machine (photo from section 3, item 3). It does not go to the PLC.
Camera power	12 - 24 VDC, up to 12 W	Copy from the old machine. If the camera has its own power supply, join its 0 V with the panel 0 V, otherwise the trigger signal has no return path.
Camera network	Ethernet switch	
Light	As on the old machine	Read the voltage on the light's label

8.2 Settings found in the saved file

From `conveyor_automation-main / backend_testing / config / camera_config / ftp-working-with-trigger-...mfa` (camera MV-ID6200M-00C-NNG, saved 23 Feb 2026). Only the values that differ from the factory settings are listed.

Setting	Value in the file	Note
IP address	192.168.0.78, mask 255.255.255.0, gateway 192.168.0.254	Fixed address. The PC program is written for this address.
Trigger mode	ON	One picture for each signal from the sensor
Trigger delay	3000000 (microseconds) = 3 seconds	Time between sensor signal and picture. Depends on belt speed and sensor position.
Exposure time / gain	100 / 150 (raw numbers in the file)	Brightness. Fine-tune with the real light.
Number of 2D codes to read	6	Up to 6 QR codes in one picture
Reading time limit	5000 ms	
Result output	TCP server, port 2002, format <code_content>; , text "NoRead" when nothing is read	The PC program connects to this port and splits the codes at the ";" sign.
Picture upload (FTP)	To 192.168.0.77, user "User", password "123456", folder <device_ip>/<image>/ with ok and ng sub-folders	The PC program is the FTP server.
FTP port	8080 in the file (21 in another saved set)	**Must be 2005.** The PC program listens on 2005, and its log of 24 Feb 2026 shows the camera connecting on 2005.

8.3 Steps

1. Mount the camera, the 12 mm lens and the light as on the old machine: same height, same angle, same sensor position.
2. **Best way:** export the settings from the old camera with IDMVS and import that file into the new camera.
3. **Other way:** import the saved file named in 8.2 into the new camera. Then check every row of the table in 8.2 and set the FTP port to 2005.
4. Save the settings in the camera (a "User Set") and make that set the one loaded at power-on. If you skip this the settings are lost when the power is switched off.
5. Test: block the sensor by hand. After the delay the light flashes and a picture is taken.

9. Network

All devices plug into the Teltonika 5-port switch. Subnet mask for all: 255.255.255.0.

Device	IP address	Where the value is set
PLC (PLC_1)	192.168.0.40	In the TIA project. Set automatically at download.
HMI (HMI_1)	192.168.0.41	In the TIA project. You also type it on the panel (section 11).
PC with the Lux WMS program	192.168.0.77	Windows / Linux network settings. Must be exactly this: the program and the camera both use it.
Camera	192.168.0.78	In the camera settings (section 8).
Laptop for programming	192.168.0.100 (any free address)	Network settings of the laptop.

From	To	Port	What
PC	PLC 192.168.0.40	502 (Modbus TCP)	Write register 40001: 0 = conveyor may run, 1 = stop. Function 06, unit 1.
PC	Camera 192.168.0.78	2002 (TCP)	The PC receives the QR codes
Camera	PC 192.168.0.77	2005 (FTP)	The camera uploads the pictures
HMI	PLC 192.168.0.40	S7 connection	HMI_Connection_1

In the PLC, register 40001 is the value **DATA1** of data block DB_CAMERA (DB5). Registers 40001 - 40010 are DATA1 - DATA10. Only DATA1 is used.

10. Load the PLC program

You need a Windows laptop with TIA Portal V21, an Ethernet cable, and the project folder **latest plc (2) / latest plc** (or the project you uploaded from the old PLC, if the reverse test in section 3 said YES).

1. Finish and check all wiring. Switch on. A new PLC shows a yellow STOP light.
2. Laptop: network adapter > IPv4 > IP 192.168.0.100, mask 255.255.255.0. Plug the laptop into the switch.
3. Open TIA Portal V21. Open project: select the file **latest plc.ap21**.
4. In the project tree click **PLC_1**. Menu **Online > Extended download to device**.
5. Type of PG/PC interface: **PN/IE**. PG/PC interface: your laptop network card. Tick **Show all compatible devices**. Click **Start search**.
6. Select the PLC in the list and click **Load**. If TIA asks to assign an IP address, click Yes / OK.
7. In the "Load preview" window accept the actions (Stop modules: "Stop all"; overwrite questions: accept). Click **Load**.

8 In "Load results" tick **Start module** and click **Finish**. The PLC light turns green (RUN).

9 Do not change anything in the program. Close without saving if TIA asks.

If TIA says the firmware does not match: the project is made for CPU firmware **V4.4**. A new CPU with V4.4 or newer accepts it. If the new CPU has older firmware, update the CPU firmware first. Do not change the CPU type in the project.

11. Load the HMI screens

1 On the KTP700 panel, at start-up press **Settings**. Open **Network Interface**: DHCP off, IP 192.168.0.41, mask 255.255.255.0.

2 Open **Transfer Settings**: switch **Enable transfer** ON. Go back to the start screen.

3 In TIA Portal click **HMI_1**. Menu **Online > Extended download to device**, interface PN/IE, your network card, **Start search**, select the panel, **Load**.

4 If TIA asks to update the panel operating system to match the project, accept. Do not switch off during the update.

5 In the preview tick "Overwrite all", click **Load**. The panel restarts and shows the project screens.

6 Check: the panel shows live values and no "#####" fields. "#####" means the HMI cannot reach the PLC.

- Screens of the old HMI (from the TIA export): AUTO, MANUAL, ALARMS, INPUT STATUS, OUTPUT STATUS, with template MAIN. The AUTO status field is green at AUTO RUNNING and red at STOPPED; the I/O lamps are red when OFF and green when ON.
- **The BACK button on OUTPUT STATUS does nothing in the old project**: it opens its own screen. If you copy the HMI exactly, use the IO STATUS button at the bottom to go back. The how-to guide (10.5) shows the one-line fix.

12. PC program (Lux WMS)

The program is in `conveyor_automation-main / conveyor_automation-main / Frontend`. The folder `backend_testing` next to it holds earlier test versions and is not needed to run the machine.

Part	File	Job
PLC bridge	backend / server.py (port 5000)	Compares the number of codes with the Stack Size. Writes 0 or 1 to PLC register 40001.
Code receiver	backend / tcp_client.py	Receives the QR codes from the camera (192.168.0.78 : 2002). Also sends each code to the company cloud server.
Picture receiver	backend / ftp_server.py (192.168.0.77 : 2005)	Receives the pictures from the camera into backend / hik-camera-data.
Picture service	backend / api_server.py (port 8000)	Gives the newest picture to the screen.
Operator screen	React web page (port 5173)	Login, orders, and the Scanning page with Start, Stop, Stack Size and Rescan Stack.

Install and start

1 Give the PC the address **192.168.0.77**, mask 255.255.255.0, on the network card that goes to the machine switch. The PC also needs internet (for example by Wi-Fi), because login and orders come from the company server.

2 Install **Python 3** and **Node.js**.

3 In the folder `Frontend/backend` run:
`pip install -r requirements.txt`

`pip install flask flask-cors requests`
The second line is needed because the code uses these three packages but the list does not name them.

4. In the folder `Frontend` run: `npm install`
5. Switch on the PLC and the camera first. Then start everything: on Linux run `./start.sh` in the Frontend folder. On Windows run `python ..\launcher.py` inside Frontend\backend, and `npm run dev` inside Frontend.
6. Open **http://localhost:5173** in the browser. Log in, choose the order, open the Scanning page. For a machine test only, the Scanning page is also reachable directly at http://localhost:5173/scan.
7. On the Scanning page: **Start** writes 0 (run), **Stop** writes 1 (stop), **Stack Size** sets how many boxes are expected.

Quick test of the PC to PLC link without the screen: one folder above Frontend, run `python3 backend_testing/test_plc_connection.py`. Type **w** then **1**: the conveyor stops. Type **w** then **0**: it runs. Type **r** to read the register back.

- If the camera was off when the program started, the code receiver stops by itself. Restart the program after the camera is on.
- If the PC does not have the address 192.168.0.77, the picture receiver cannot start.

13. Test step by step

Do the tests in this order. Do not go to the next step until the present one is correct.

#	Test	Correct result
1	Power off. Check with a multimeter: no short between +24 V and 0 V, and none between L and N.	No continuity
2	Switch on isolator and MCBs, with the VFD MCB (C10) still OFF.	24.0 V on the terminal row. PLC, HMI and switch powered.
3	Look at the PLC input LEDs with nothing pressed.	I0.3 ON, I0.4 ON, I0.5 OFF, I0.0 OFF
4	Turn the selector to AUTO, then MANUAL.	I0.1 ON in AUTO. Amber lamp ON in MANUAL.
5	Press START, then STOP, then Emergency (one at a time).	I0.2 ON while pressed. I0.3 OFF while pressed. I0.4 OFF while Emergency is pushed in.
6	After pushing Emergency: release it and press FAULT RESET on the HMI.	Red lamp ON with the alarm text on the HMI. After reset the red lamp goes OFF.
7	Switch ON the VFD MCB. Enter the VFD parameters (section 7).	VFD display normal, no fault code. VFD healthy lamp ON.
8	Selector MANUAL. Hold FWD JOG on the HMI (or the green START button).	Q0.0 ON, conveyor runs forward. If the direction is wrong, power off and swap two motor wires (U and V).
9	Hold REV JOG on the HMI.	Q0.1 ON, conveyor runs in reverse.
9b	Only if you added the red-button reverse (how-to guide part 12): selector MANUAL, hold the red button, then release.	Q0.1 ON while held, conveyor reverses; stops on release. In AUTO the red button still stops the cycle.
10	Selector AUTO. Press START. The PC program is not needed for this step.	Green lamp ON, conveyor runs forward.
11	On the PC write 1 to register 40001 (Stop button, or the quick test of section 12). Then write 0.	Conveyor stops at 1 (green lamp stays ON) and runs again at 0.
12	Press STOP.	Conveyor stops, green lamp OFF.

#	Test	Correct result
13	Start again, then press Emergency while running.	Conveyor stops, red lamp ON.
14	Camera: block the sensor by hand.	After the trigger delay a picture is taken. The PC screen shows the picture and the codes.
15	Full test: run a correct row of boxes, then a row with one box missing.	Correct row: conveyor keeps running. Missing box: conveyor stops and the PC shows a mismatch. Rescan Stack: it runs again.

14. Good to know

- **Old sensor logic is still inside the PLC program, switched off.** The program contains a cycle "sensor, delay, stop, wait for camera". It is disabled by a contact that never passes. That is normal for this machine, because the sensor now goes to the camera.
- **If the PC stops the belt, the green START button does not restart it.** Use Rescan Stack or Start on the PC, which writes 0 again.
- **The HMI shows "AUTO RUNNING" also while the PC has stopped the belt.** The HMI text only looks at the START / STOP state.
- **When the PLC is switched on, register 40001 is 0.** So START runs the conveyor even when the PC program is not running. Nothing checks the boxes then.
- **Network.** Any device on this network can write the register, and can also switch PLC outputs through Modbus. Keep the machine network separate from the office network.
- **Alarms in the program:** "EMERGENCY PB PRESSED FAULT (I0.4)" and "CONVEYOR VFD FAULT (I0.5)". Both stay stored until FAULT RESET is pressed on the HMI.
- **The PC cannot start the cycle.** It can only stop the belt (1) and let it run again (0). START on the panel, in AUTO, is always needed first.
- **In MANUAL the red button does nothing** in the old program. Reverse comes only from the HMI REV JOG button, unless you add part 12 of the how-to guide.

Appendix A. Program reference

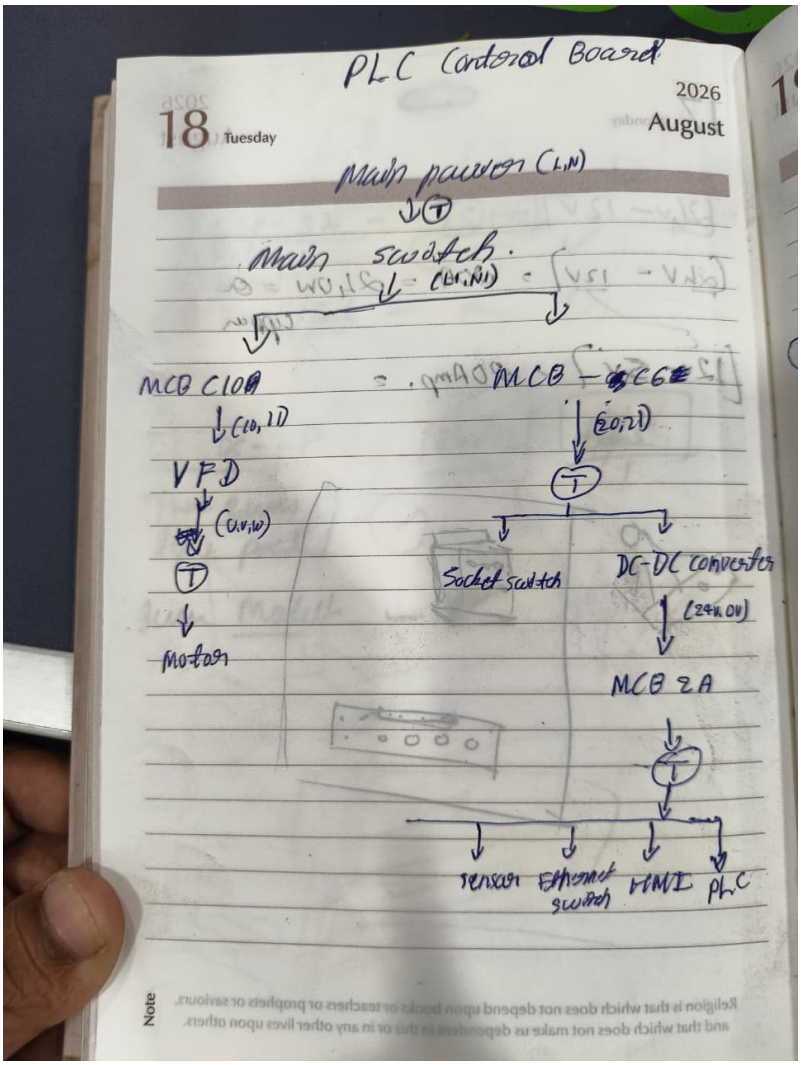
Block	Job
Main (OB1)	Runs the six blocks below, in this order, again and again
FC_AUTO	6 networks. Working: 1 start / stop memory; 3 (first row) conveyor runs while started and register 40001 = 0; 6 status number for the HMI (10 running, 20 stopped). Networks 2, 4, 5 and the rest of 3 are the old sensor cycle, switched off by a normally closed AlwaysTRUE contact
FC_ALARMS	Stores Emergency and VFD faults, makes the common FAULT signal
FC_GENERAL	Emergency healthy flag, Auto / Manual mode, fault reset from the HMI
FC_MANUAL	Jog in manual mode while held: forward from the HMI FWD JOG or the green START; reverse from the HMI REV JOG only
FC_OUTPUTS	Drives Q0.0 - Q0.5
FC_TCP_IP	Modbus TCP server on port 502, counts good and bad messages
DB_HMI (DB1), DB_ALARMS (DB2), DB_AUTO (DB3), DB_CAMERA (DB5), DB_TCP_IP (DB12), MB_SERVER_DB (DB4)	Data used by the blocks, the HMI and Modbus

Tags: three PLC tag tables (DI, DO, Default). Clock memory byte ON at MB0, system memory byte at MB1. DB_TCP_IP is optimized; the other four DBs are not. The exact program, network by network, is in the folder **07_Old_Program_Export** (OLD_PROGRAM_LOGIC.txt and OLD_HMI.txt) and in README

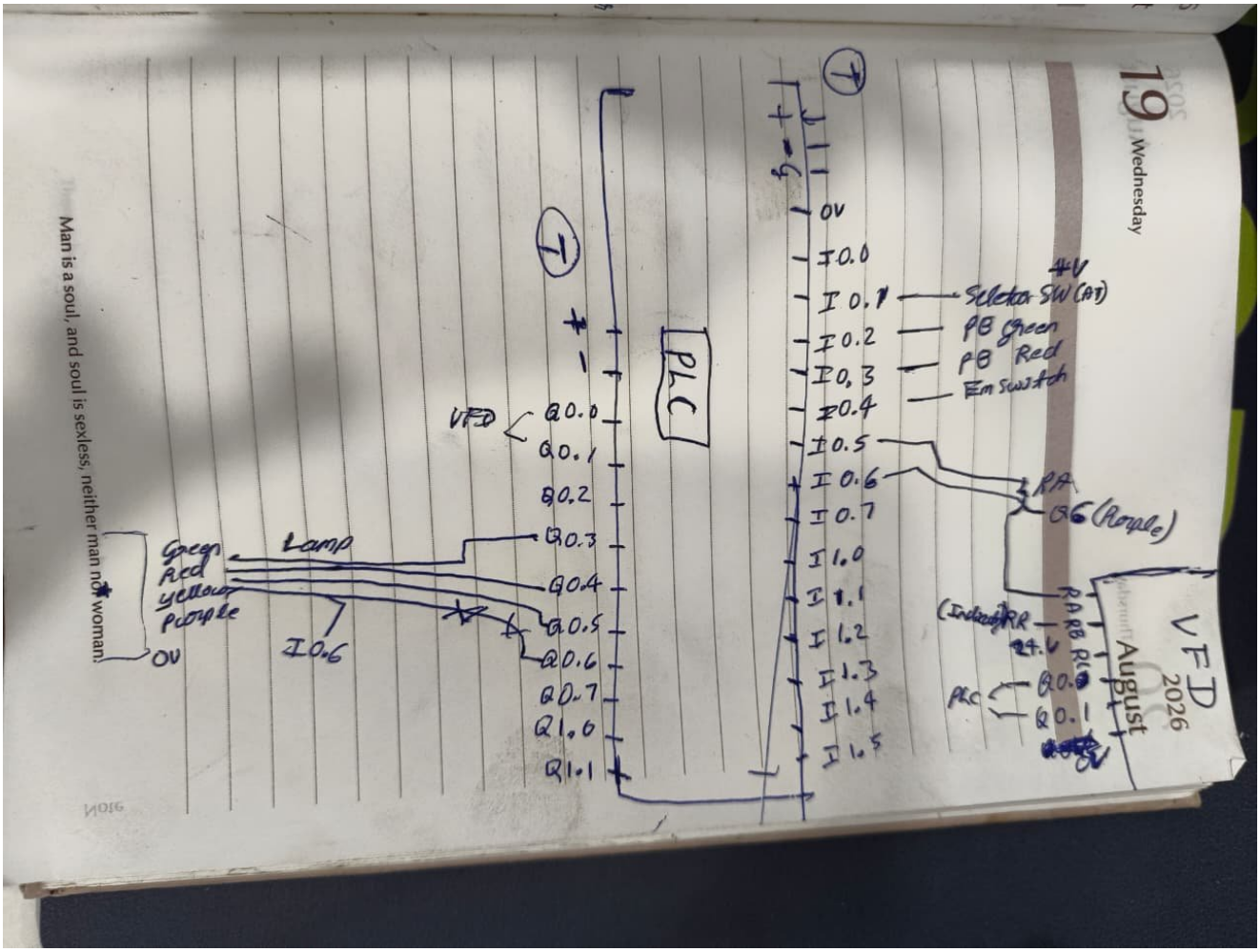
section 11; the how-to guide (Revision 3) draws every network exactly.

Appendix B. Your hand drawings

Power distribution (18 Aug page)



PLC inputs / outputs and VFD relay (19 Aug page)



Appendix C. Every non-zero setting of the old drive

Read from the old drive (VFD004EL21W-1) on 7 October 2026. Names, units and factory values are from the Delta VFD-EL user manual. **Value** is what the keypad shows; **raw** is the number the laptop tool prints. All parameters not listed here are 0 on the old drive (full list: 04_VFD/VFD_OLD_MACHINE/old_vfd_parameters.txt).

No.	Name	Value	Raw	Factory (manual)	Note
00.00	Identity code of the drive	2	2	-	read only
00.01	Rated current of the drive	2.5 A	25	-	read only
00.05	User-defined coefficient K	1.0	10	1.0	= factory.
00.06	Software version	2.06	206	-	read only
00.07	(not in the VFD-EL manual)	4082	4082	-	EL-W only; leave
00.11	(reserved in the VFD-EL manual)	1	1	-	EL-W only; compare with the new drive
00.12	50 Hz base voltage selection	1	1	0	**DIFFERENT from factory.** 0 = 230 V / 400 V, 1 = 220 V / 380 V
01.00	Maximum output frequency	70.00 Hz	7000	60.00	**DIFFERENT from factory.**
01.01	Maximum voltage frequency (motor rated frequency)	50.00 Hz	5000	60.00	**DIFFERENT from factory.**
01.02	Maximum output voltage (motor rated voltage)	220.0 V	2200	220.0	= factory.
01.03	Mid-point frequency	1.50 Hz	150	1.50	= factory.
01.04	Mid-point voltage	10.0 V	100	10.0	= factory.
01.05	Minimum output frequency	1.50 Hz	150	1.50	= factory.
01.06	Minimum output voltage	10.0 V	100	10.0	= factory.
01.07	Output frequency upper limit	110.0 %	1100	110.0	= factory.
01.09	Acceleration time 1	5.0 s	50	10.0	**DIFFERENT from factory.**
01.10	Deceleration time 1	0.5 s	5	10.0	**DIFFERENT from factory.**
01.11	Acceleration time 2	10.0 s	100	10.0	= factory. not used (no input selects time 2)
01.12	Deceleration time 2	0.1 s	1	10.0	**DIFFERENT from factory.** not used (no input selects time 2)
01.13	Jog acceleration time	0.1 s	1	1.0	**DIFFERENT from factory.** drive jog function, not used
01.14	Jog deceleration time	1.0 s	10	1.0	= factory. drive jog function, not used
01.15	Jog frequency	6.00 Hz	600	6.00	= factory. drive jog function, not used
01.21	Simple positioning stop frequency 1	5.00 Hz	500	5.00	= factory. not used
01.22	Simple positioning stop frequency 2	10.00 Hz	1000	10.00	= factory. not used
01.23	Simple positioning stop frequency 3	20.00 Hz	2000	20.00	= factory. not used
01.24	Simple positioning stop frequency 4	30.00 Hz	3000	30.00	= factory. not used

No.	Name	Value	Raw	Factory (manual)	Note
01.25	Simple positioning stop frequency 5	40.00 Hz	4000	40.00	= factory. not used
01.26	Simple positioning stop frequency 6	50.00 Hz	5000	50.00	= factory. not used
01.27	Simple positioning stop frequency 7	60.00 Hz	6000	60.00	= factory. not used
02.00	Source of the speed (frequency) command	4	4	1	**DIFFERENT from factory.** 0 = keypad UP / DOWN keys, 1 = 0-10 V on AVI, 2 = 4-20 mA on ACI, 3 = RS-485, 4 = keypad potentiometer (knob)
02.01	Source of the run command	1	1	1	= factory. 0 = keypad, 1 = external terminals (keypad STOP works), 2 = external terminals (keypad STOP off)
02.03	PWM carrier frequency	8 kHz	8	8	= factory.
02.05	Run at power-on	1	1	1	= factory. 1 = do not run by itself when power comes on
02.06	Loss of ACI signal	1	1	1	= factory. analog input not used
02.08	UP / DOWN rate	0.01 Hz	1	0.01	= factory. not used
02.11	Keypad frequency command	50.00 Hz	5000	60.00	**DIFFERENT from factory.** not used (speed comes from the knob)
02.12	Communication frequency command	50.00 Hz	5000	60.00	**DIFFERENT from factory.** not used
02.15	Initial frequency setpoint	60.00 Hz	6000	60.00	= factory. not used
02.16	Display: master frequency source	1	1	-	read only. 1 = first source (02.00)
02.17	Display: operation command source	4	4	-	read only. 4 = external terminals
03.00	Function of the relay RA / RB / RC	8	8	8	= factory. 8 = fault indication
03.04	Analog output gain	0 %	0	100	**DIFFERENT from factory.** analog output not used
03.08	Cooling fan control	3	3	0	**DIFFERENT from factory.** 0 = always on, 3 = on when the heat sink is warm
03.13	Display: status of the relay	1	1	-	read only
04.02	Keypad potentiometer gain	100.0 %	1000	100.0	= factory.
04.05	Function of input MI3	0	0	1	**DIFFERENT from factory.** 0 = no function, 1 = multi-step speed 1
04.06	Function of input MI4	0	0	2	**DIFFERENT from factory.** 0 = no function, 2 = multi-step speed 2
04.07	Function of input MI5	5	5	3	**DIFFERENT from factory.** 5 = external reset, 3 = multi-step speed 3

No.	Name	Value	Raw	Factory (manual)	Note
04.08	Function of input MI6	14	14	4	**DIFFERENT from factory.** 14 = external fault (E.F.) input, 4 = multi-step speed 4
04.10	Digital input debouncing time	1 x 2 ms	1	1	= factory.
04.13	Maximum AVI voltage	see raw	100	10.00	analog input not used
04.14	Maximum AVI frequency	100.0 %	1000	100.0	= factory. analog input not used
04.15	Minimum ACI current	4.0 mA	40	4.0	= factory. analog input not used
04.17	Maximum ACI current	20.0 mA	200	20.0	= factory. analog input not used
04.18	Maximum ACI frequency	100.0 %	1000	100.0	= factory. analog input not used
04.19	Start / stop control mode (VFD-EL-W)	1	1	0	**DIFFERENT from factory.** 0 = RUN/STOP by MI1 only; 1 = 2-wire / 3-wire with MI1, MI2, MI3. Must be 1 (MI2 = reverse)
04.26	Display: status of the MI inputs	63	63	-	read only
06.00	Over-voltage stall prevention	390.0 V	3900	390.0	= factory.
06.01	Over-current stall prevention during acceleration	170 %	170	170	= factory.
06.02	Over-current stall prevention during operation	170 %	170	170	= factory.
06.04	Over-torque detection level	150 %	150	150	= factory.
06.05	Over-torque detection time	0.1 s	1	0.1	= factory.
06.06	Electronic thermal overload relay (motor protection)	2	2	2	= factory. 0 = standard motor, 1 = special motor, 2 = disabled
06.07	Electronic thermal characteristic	60 s	60	60	= factory.
06.14	(not in the VFD-EL manual)	30	30	-	EL-W only; compare with the new drive
06.15	(not in the VFD-EL manual)	44	44	-	EL-W only; compare with the new drive
06.16	(not in the VFD-EL manual)	30	30	-	EL-W only; compare with the new drive
06.19	(not in the VFD-EL manual)	8	8	-	EL-W only; compare with the new drive
07.00	Motor rated current	2.5 A	25	= drive rated current (2.5)	
07.01	Motor no-load current	1.0 A	10	0.4 x rated (1.0)	**DIFFERENT from factory.**
07.06	Motor rated slip	3.00 Hz	300	3.00	= factory.
07.07	Slip compensation limit	200 %	200	200	= factory.
07.08	Torque compensation time constant	0.10 s	10	0.30	**DIFFERENT from factory.** compare with the new drive
07.09	Slip compensation time constant	0.20 s	20	0.20	= factory.
07.10	Accumulated motor run time (minutes)	633 min	633	-	a counter, not a setting
07.13	PTC input debouncing time	100 x 2 ms	100	100	= factory. PTC not used

No.	Name	Value	Raw	Factory (manual)	Note
07.14	Motor PTC overheat protection level	2.4 V	24	2.4	= factory. PTC not used
07.15	Motor PTC overheat warning level	1.2 V	12	1.2	= factory. PTC not used
07.16	Motor PTC overheat reset delta level	0.6 V	6	0.6	= factory. PTC not used
08.05	Maximum allowable power loss time	2.0 s	20	2.0	= factory.
08.06	Base-block speed search	1	1	1	= factory.
08.07	B.B. time for speed search	0.5 s	5	0.5	= factory.
08.08	Current limit for speed search	150 %	150	150	= factory.
08.16	Auto reset time at restart after fault	60.0 s	600	60.0	= factory. auto restart itself is off (08.15 = 0)
08.24	Fire mode frequency	60.00 Hz	6000	60.00	= factory. fire mode off
09.00	Communication address	1	1	1	= factory.
09.01	Communication speed	1	1	1	= factory. 1 = 9600 baud
09.02	Transmission fault treatment	3	3	3	= factory. 3 = no warning, keep operating
09.07	Response delay time	1 x 2 ms	1	1	= factory.
10.02	PID parameter	10	10	-	PID is switched off (10.00 = 0); the values of group 10 have no effect
10.03	PID parameter	100	100	-	as above
10.05	PID parameter	100	100	-	as above
10.07	PID parameter	100	100	-	as above
10.08	PID parameter	600	600	-	as above
10.10	PID parameter	100	100	-	as above
10.12	PID parameter	100	100	-	as above
10.13	PID parameter	50	50	-	as above
10.18	PID parameter	999	999	-	as above
10.21	PID parameter	60	60	-	as above
10.23	PID parameter	10	10	-	as above
10.26	PID parameter	5	5	-	as above
10.37	PID parameter	60	60	-	as above
10.38	PID parameter	6000	6000	-	as above
10.39	PID parameter	10	10	-	as above
10.40	PID parameter	4800	4800	-	as above
10.41	PID parameter	10	10	-	as above
10.43	PID parameter	1	1	-	as above
10.45	PID parameter	600	600	-	as above

Group 12 is the drive's own table of how many parameters each group has; it is not a setting.